Spectral Unmixing of Hyperspectral Images Using Deep Learning Techniques

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Problem Statement

OBJECTIVE: To develop an efficient deep learning framework for accurate mineral classification and abundance estimation from hyperspectral imagery.

DESCRIPTION:

- Mineral mapping is critical for geological exploration and mining applications.
- Hyperspectral imagery provides rich spectral information for mineral identification.
- High dimensionality and spectral similarity between minerals make accurate classification challenging.
- Robust spectral-spatial learning methods are required for reliable mineral classification and abundance estimation.

What is Spectral Unmixing?

DEFINITION:
Spectral unmixing decomposes a mixed pixel's spectral response into pure material signatures (endmembers) and their fractional contributions (abundances).

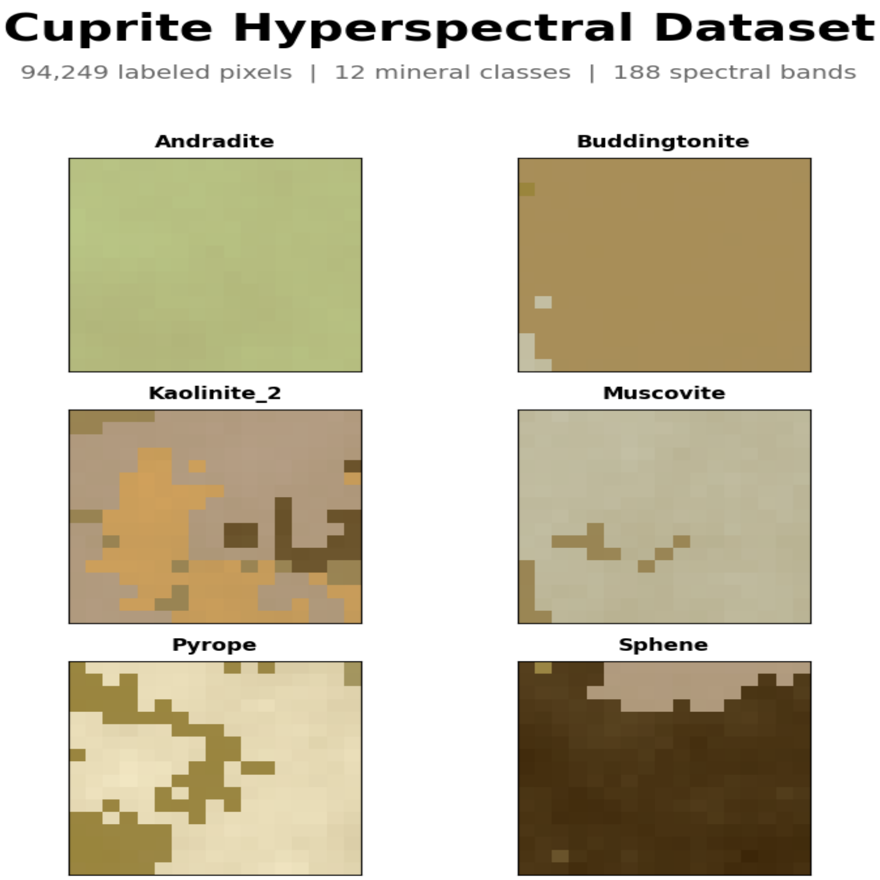
Challenges:
Most pixels contain multiple minerals mixed together
Spectra of similar minerals overlap significantly
Sensor noise and atmospheric effects distort signals

Goal: Given a mixed pixel spectrum → identify which minerals are present and in what proportion

Baseline Methods

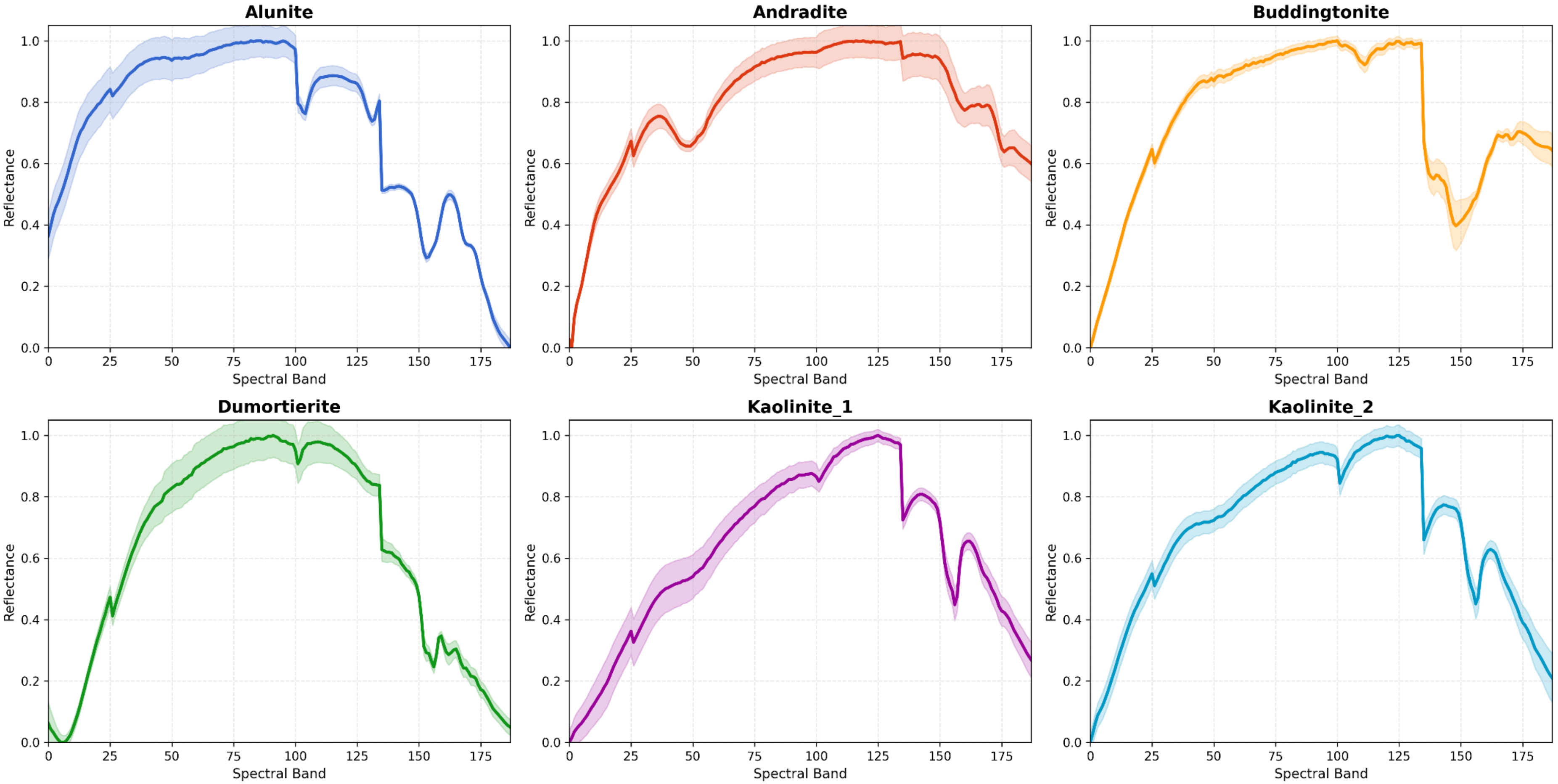
Model	Core Idea	Limitation
CNNAE (Convolutional Neural Network Autoencoder Unmixing) [1]	CNN encoder-decoder learns spectral-spatial features for abundance estimation	Limited ability to capture complex feature relationships
DAEN (Deep Autoencoder Network) [2]	Stacked autoencoders extract compact latent spectral representations	Important spectral structures may be lost during compression
VAE-Based [3]	Probabilistic latent space models endmember variability and spectral uncertainty	Limited preservation of spatial neighborhood information
DUENet (Denoising Unmixing Encoder Network) [4]	3D CNN + VAE + DTW neurons + dynamic capsule routing for joint unmixing and denoising	DTW-based dynamic routing causes gradient stagnation during training

Dataset [5]

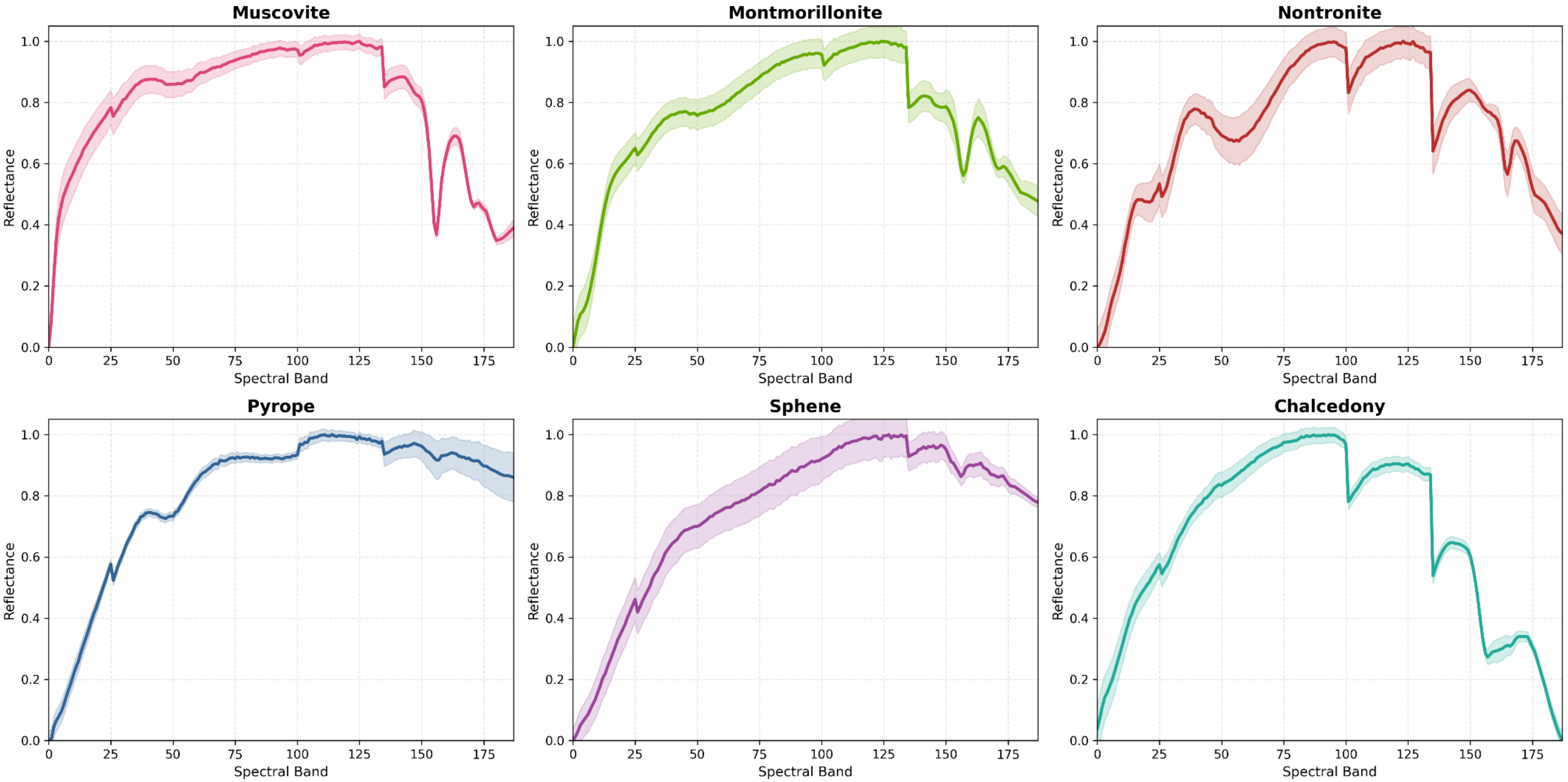


Sample 17×17 spectral-spatial patches extracted for each mineral class

Dataset – Spectral Signatures



Dataset – Spectral Signatures



Methodology Overview - DUENet

Proposed Methodology – DUENet for Spectral Unmixing

Input Hyperspectral Image (HSI)

- Acquire hyperspectral image cube containing spectral information across multiple wavelength bands

Patch Extraction and Data Preparation

- Extract local spatial neighborhoods using fixed-size patches (7 × 7)
- Generate labeled samples using dominant mineral abundance information

Spectral–Spatial Feature Extraction (3D CNN)

- Learn joint spectral and spatial characteristics using 3D convolutions with Gaussian Interpolation

$$F = f(X)$$

where
X= input HSI patch
F= extracted feature representation

Variational Latent Representation Learning

- Compress features into compact latent representation

$$z = \mu + \sigma\epsilon$$

where Z= latent vector

Methodology Overview

Primary Capsule Generation

- Convert latent vectors into capsule representations

$$u_i = W_i z$$

Dynamic Routing Between Capsules

- Route feature information toward mineral-specific capsules

$$v_j = squash\left(\sum_i c_{ij}\, \hat{u}_{j|i}\right)$$

Mineral Abundance Estimation

- Estimate fractional mineral abundance from capsule vector length

$$A_j = || v_j ||$$

Methodology Overview

Spectral Reconstruction and Optimization

- Reconstruct spectral signatures and minimize reconstruction error

$$L = \gamma_1(L_G + L_S + L_{SD}) + \gamma_2 L_C$$

where

$$L_G = -\frac{1}{2}(1 + \log\sigma^2 - \mu^2 - \sigma^2)$$

$$L_S = \cos^{-1}\left(\frac{x \cdot \hat{x}}{\| x \|\| \hat{x} \|}\right)$$

$$L_S = \cos^{-1}\left(\frac{x \cdot \hat{x}}{\| x \|\| \hat{x} \|}\right)$$

$$L_{SD} = 1 - SSIM(x, \hat{x})$$

where

$$SSIM = \frac{(2\mu_x\mu_r + C_1)(2\sigma_{xr} + C_2)}{(\mu_x^2 + \mu_r^2 + C_1)(\sigma_x + \sigma_r + C_2)}$$

Output

- Mineral abundance maps
- Reconstructed spectral signatures

Architecture — DUENet

Inputs: 7×7 hyperspectral patches → transpose → (B,188,7,7))

3D CNN Encoder: Apply spectral–spatial convolutions using two Conv3D blocks
- Conv Block 1: 1 → 16 channels
- Conv Block 2: 16 → 32 channels
- Learn local spectral-spatial representations

Spectral Pooling: Reduce redundant spectral dimensions
188 bands → ~16 feature bands

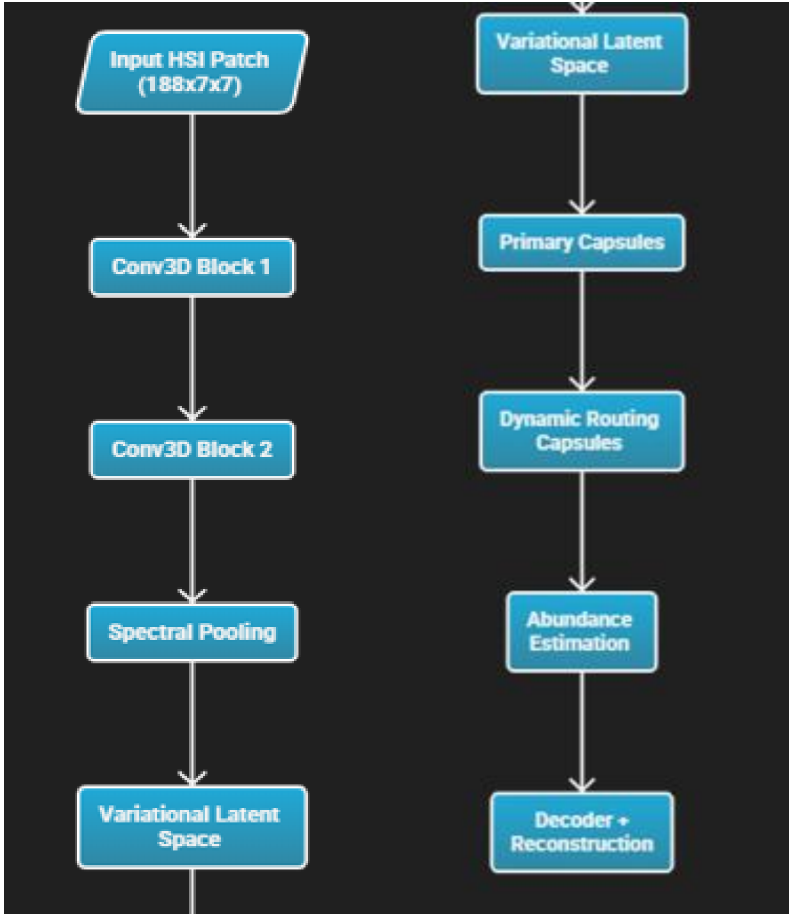
Variational Latent Layer: Extract latent representation
Generate: $\mu, \log\sigma^2$
Reparameterization: $z = \mu + \sigma\epsilon$
Output: (B, 64)

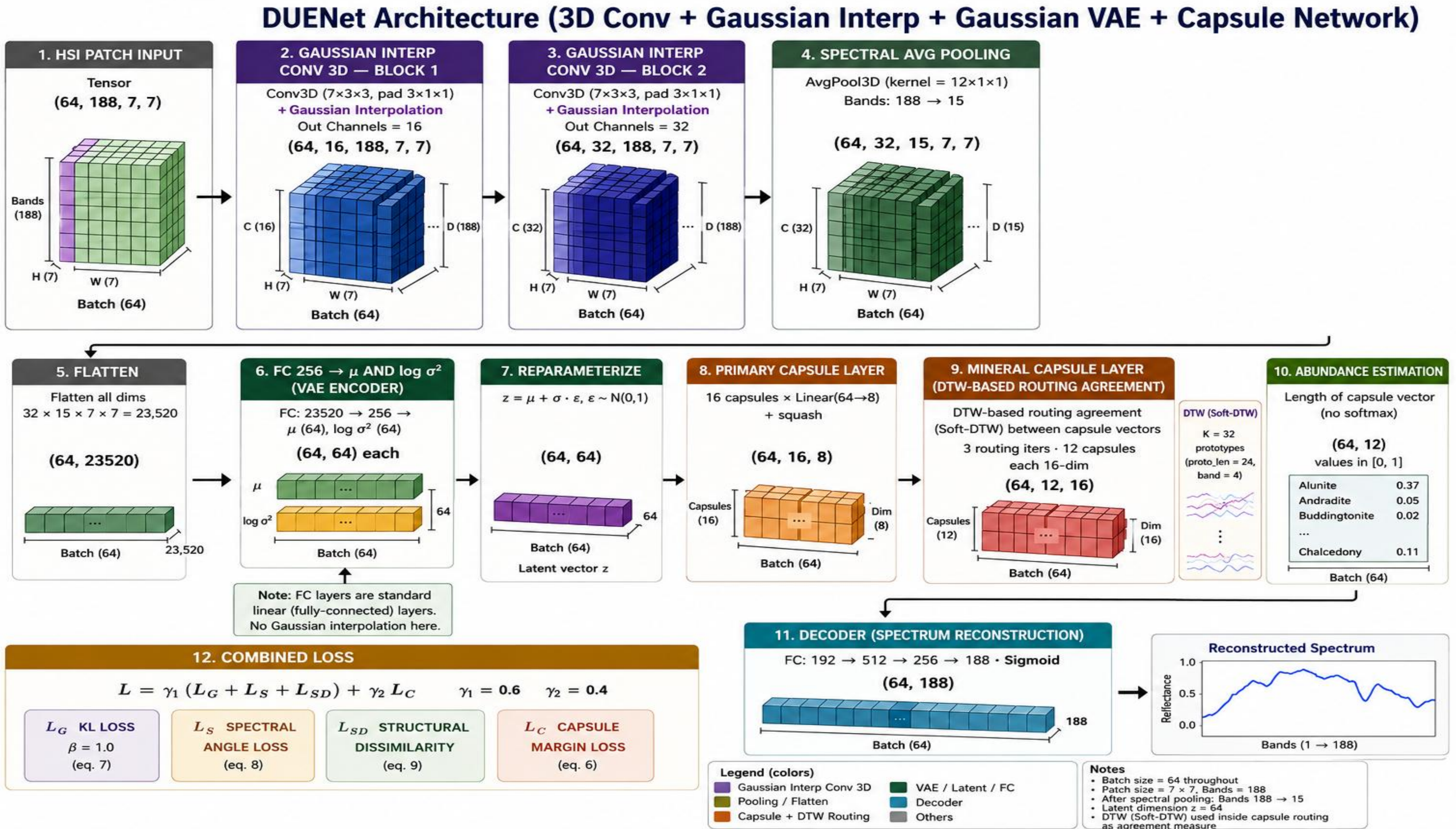
Primary Capsule Layer: Latent vector → 16 capsule vectors; 64 → (16 × 8)

Dynamic Routing Capsule Layer: Route lower-level capsules into mineral capsules; (16×8) → (12×16)

Abundance Estimation: Capsule vector length → mineral abundance fractions

Decoder + Reconstruction: Flatten selected capsule vectors → Fully Connected layers





DUENet Architecture (3D Conv + Gaussian Interp + Gaussian VAE + Capsule Network)
1. HSI PATCH INPUT
Tensor
(64, 188, 7, 7)
Bands (188)
H (7)
W (7)
Batch (64)
2. GAUSSIAN INTERP CONV 3D — BLOCK 1
Conv3D (7×3×3, pad 3×1×1)
+ Gaussian Interpolation
Out Channels = 16
(64, 16, 188, 7, 7)
C (16)
D (188)
H (7)
W (7)
Batch (64)
3. GAUSSIAN INTERP CONV 3D — BLOCK 2
Conv3D (7×3×3, pad 3×1×1)
+ Gaussian Interpolation
Out Channels = 32
(64, 32, 188, 7, 7)
C (32)
D (188)
H (7)
W (7)
Batch (64)
4. SPECTRAL AVG POOLING
AvgPool3D (kernel = 12×1×1)
Bands: 188 → 15
(64, 32, 15, 7, 7)
C (32)
D (15)
H (7)
W (7)
Batch (64)
5. FLATTEN
Flatten all dims
32 × 15 × 7 × 7 = 23,520
(64, 23520)
Batch (64)
23,520
6. FC 256 → μ AND log σ²
(VAE ENCODER)
FC: 23520 → 256 →
μ (64), log σ² (64)
(64, 64) each
μ
log σ²
64
Batch (64)
Note: FC layers are standard linear (fully-connected) layers. No Gaussian interpolation here.
7. REPARAMETERIZE
z = μ + σ · ε, ε ~ N(0,1)
(64, 64)
64
Batch (64)
Latent vector z
8. PRIMARY CAPSULE LAYER
16 capsules × Linear(64→8)
+ squash
(64, 16, 8)
Capsules (16)
Dim (8)
Batch (64)
9. MINERAL CAPSULE LAYER
(DTW-BASED ROUTING AGREEMENT)
DTW-based routing agreement
(Soft-DTW) between capsule vectors
3 routing iters · 12 capsules
each 16-dim
(64, 12, 16)
Capsules (12)
Dim (16)
Batch (64)
DTW (Soft-DTW)
K = 32
prototypes
(proto_len = 24,
band = 4)
10. ABUNDANCE ESTIMATION
Length of capsule vector
(no softmax)
(64, 12)
values in [0, 1]
Alunite 0.37
Andradite 0.05
Buddingtonite 0.02
...
Chalcedony 0.11
Batch (64)
11. DECODER (SPECTRUM RECONSTRUCTION)
FC: 192 → 512 → 256 → 188 · Sigmoid
(64, 188)
188
Batch (64)
Reconstructed Spectrum
Reflectance
1.0
0.5
0.0
Bands (1 → 188)
12. COMBINED LOSS
$L = \gamma_1 (L_G + L_S + L_{SD}) + \gamma_2 L_C$ $\gamma_1 = 0.6$ $\gamma_2 = 0.4$
L_G KL LOSS
$\beta = 1.0$
(eq. 7)
L_S SPECTRAL ANGLE LOSS
(eq. 8)
L_{SD} STRUCTURAL DISSIMILARITY
(eq. 9)
L_C CAPSULE MARGIN LOSS
(eq. 6)
Legend (colors)
Gaussian Interp Conv 3D
Pooling / Flatten
Capsule + DTW Routing
VAE / Latent / FC
Decoder
Others
Notes
• Batch size = 64 throughout
• Patch size = 7 × 7, Bands = 188
• After spectral pooling: Bands 188 → 15
• Latent dimension z = 64
• DTW (Soft-DTW) used inside capsule routing as agreement measure

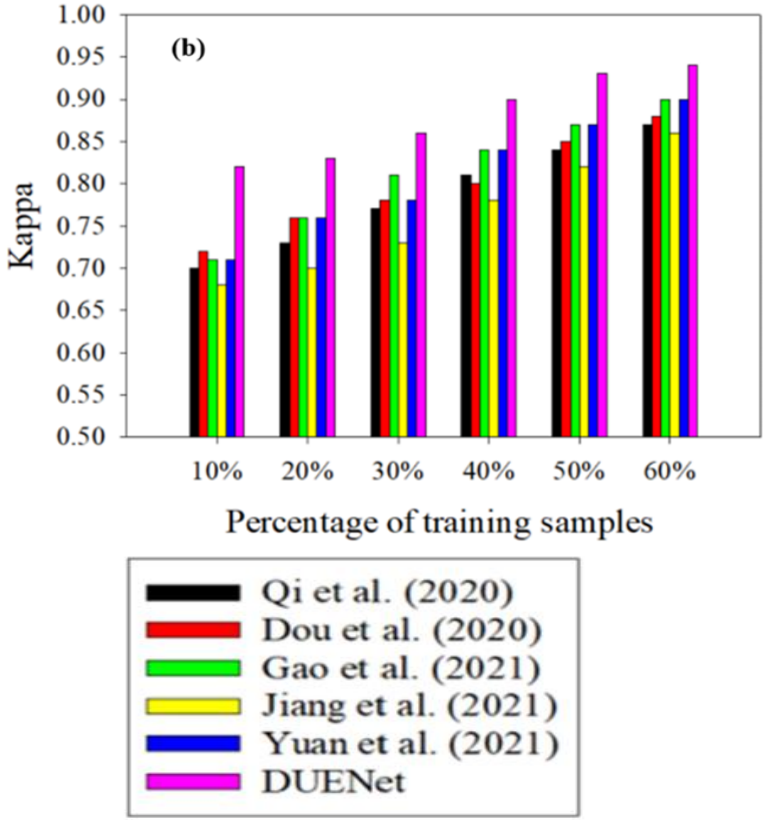
Reported Results from Original DUENet Paper

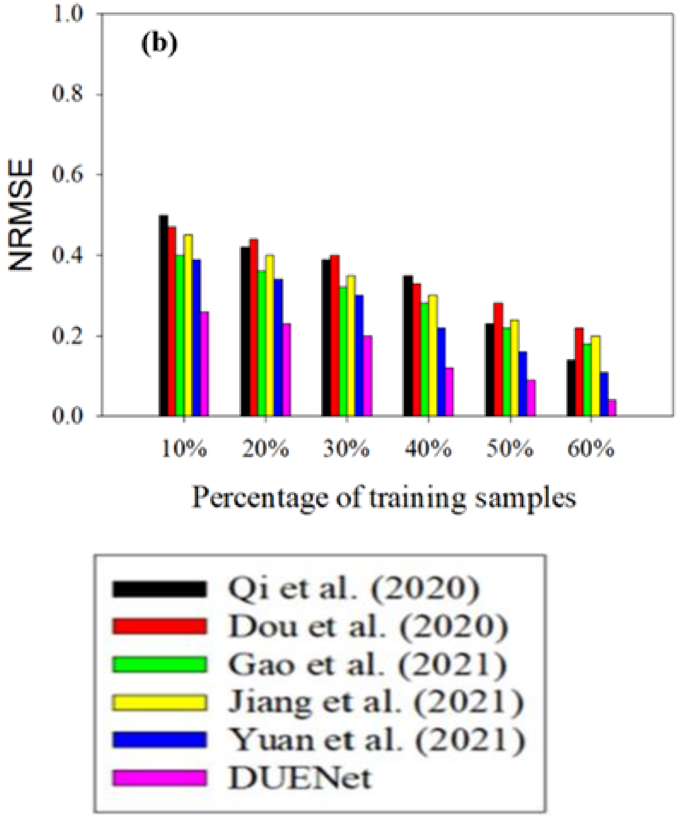
Table 3
Comparison of denoising unmixing encoder network (DUENet) with benchmark deep learning (DL) classifiers for 70% of training samples*.

Dataset	Benchmark approaches	NRMSE	Overall Accuracy	Kappa
Cuprite Dataset	(Su et al., 2019) (Su et al., 2019)	0.63	90.13	0.87
	(Ozkan et al., 2019)	0.40	92.80	0.88
	(Borsoi et al., 2020)	0.51	92.92	0.89
	(Palsson et al., 2019)	0.47	91.08	0.89
	(Qi et al., 2020)	0.35	93.08	0.89
	(Qian et al., 2020)	0.42	93.19	0.90
	(Dou et al., 2020)	0.33	94.15	0.91
	(Jiang et al., 2021)	0.30	95.32	0.92
	(Gao et al., 2021)	0.28	96.14	0.93
	(Yuan et al., 2021)	0.22	96.07	0.93
	DUENet	**0.12**	**97.56**	**0.95**

Reported Results from Original DUENet Paper

Accuracy analysis of denoising unmixing encoder network (DUENet) and benchmark unmixing approaches for the Cuprite dataset.





Motivation for Modifications

Problem 1: Dynamic routing collapses at initialization
All capsule weights start near zero → all DTW costs between prediction vectors are approximately equal → routing logits update uniformly → softmax stays uniform → model predicts the majority class for every pixel
Result: accuracy stuck at ~8% (random chance for 12 classes) for the first several epochs
Fix: replace iterative routing with a single learned linear projection — gradients flow directly from the loss to the weights from step 1

Problem 2: KL collapse
With a large entanglement penalty (β=4) and high learning rate, the KL loss dominates in epoch 1 — the encoder immediately pushes mu→0, log_var→0 regardless of input
Result: every pixel maps to the same latent code — the encoder stops learning
Fix: KL warmup starting from β=0, gradually increasing to β=0.1 over 8 epochs

Problem 3: DTW location was wrong for gradient flow
DTW neurons operating on near-zero feature vectors produce near-zero costs for all inputs — no discrimination signal
Fix: move DTW to after the conv blocks where features are already nonzero, use it as a spectral gating mechanism instead

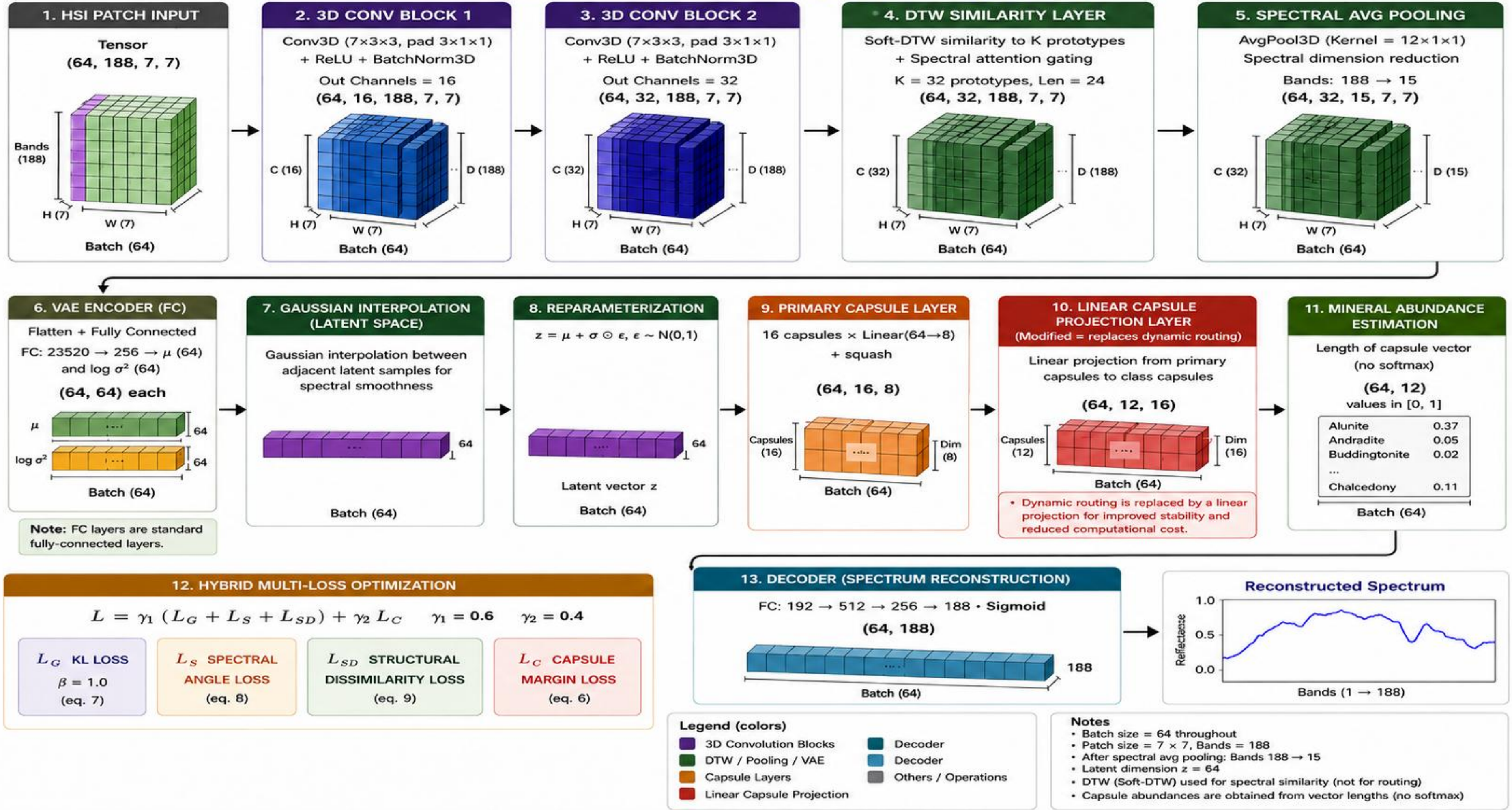
Modified Proposed Model – SpectralNet

Component	Original	Modified	Reason
Capsule routing	Iterative dynamic routing (3 iterations, DTW agreement)	Single learned linear projection	The voting algorithm produces near-zero gradients at initialization so the model never learns. Direct mapping allows gradients to flow from the very first step.
DTW placement	Inside neuron activations + capsule routing agreement	Feature gating layer after conv blocks, before pooling	Operates on nonzero conv features — avoids near-zero initialization problem
DTW prototypes	8 learnable prototypes, length 12	32 prototypes, length 24	Larger latent space gives the model more room to separately encode the 12 different minerals without them overlapping
Latent dim	32	64	Better spectral manifold representation and class separability
Capsule dims	Primary 4D, Class 8D	Primary 8D, Class 16D	Stronger abundance-aware feature encoding

Modified Proposed Model - SpectralNet

Component	Original	Modified	Reason
KL loss	Constant β=4	Warmup from β=0 → 0.1 over 8 epochs	Prevents posterior collapse in early training
LR schedule	Step decay ×0.5 every 100 epochs	Cosine annealing to 5% of peak	Smoother convergence, avoids overshooting late in training
Gaussian smoothing	Applied to convolution kernel positions (input smoothing)	Applied to VAE latent vector mu and log_var	Enforces smoothness prior on latent space — nearby dims represent similar spectral characteristics
Classification loss	Cross-entropy + L1 latent reg	Capsule margin loss	Cross entropy on near-zero capsule outputs at initialization causes unstable gradients. Margin loss has smooth bounded gradients regardless of the output magnitude

SpectralNet Architecture (3D Conv + DTW Similarity + Gaussian Interp + Gaussian VAE + Linear Capsule Projection)

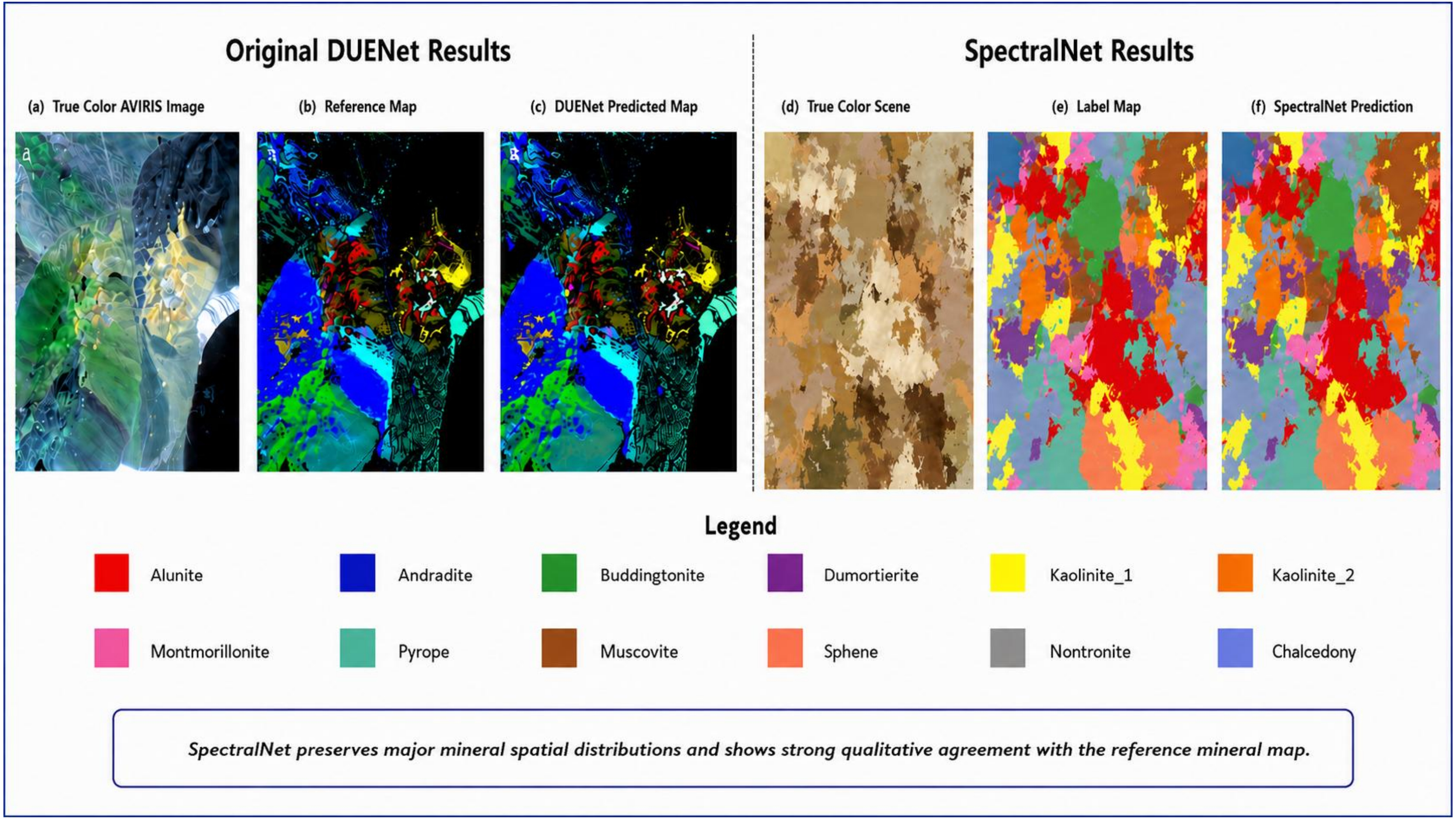


Comparative Results - Quantitative

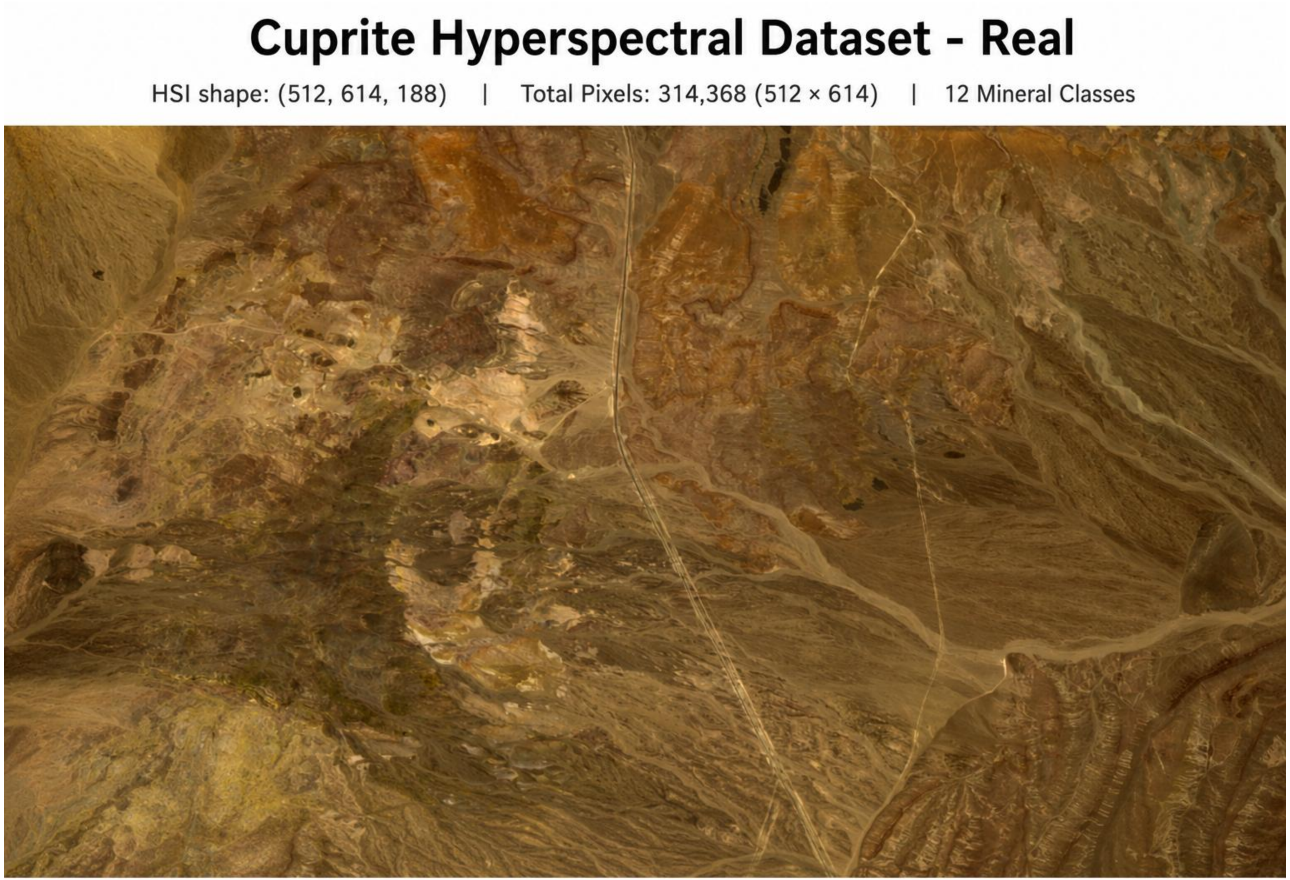
Model	**Split**	**Patch Size**	**NRMSE ↓**	**Overall Accuracy ↑**	**Kappa ↑**	**Notes**
DUENet (Pattathal et al. 2022)	Random 70/30	—	0.12	97.56%	0.95	Paper reported
SpectralNet	Random 70/30	7×7	0.039	99.86%	0.9985	Epoch 2
SpectralNet	Random 70/30	13×13	0.064	99.08%	0.9898	Epoch 2
SpectralNet	Random 70/30	17×17	0.083	98.06%	0.9784	Epoch 3
SpectralNet	**Spatial split**	7×7	0.057	99.14%	~0.99	Spatially unseen region — no patch overlap

Spatial split: top 70% of rows for training, bottom 30% for testing with a gap to prevent patch overlap. Tests generalization to unseen spatial regions.

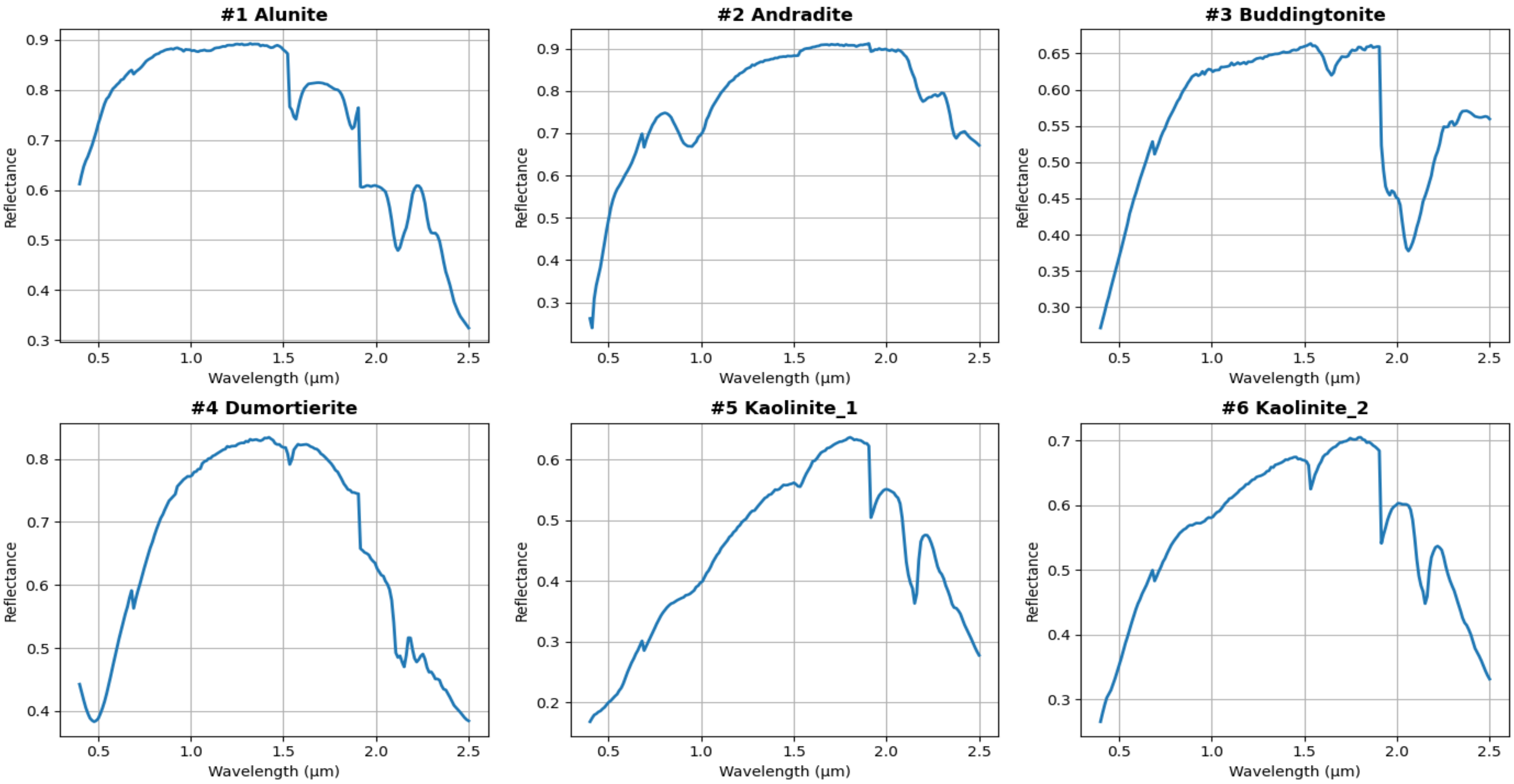
Comparative Results - Qualitative



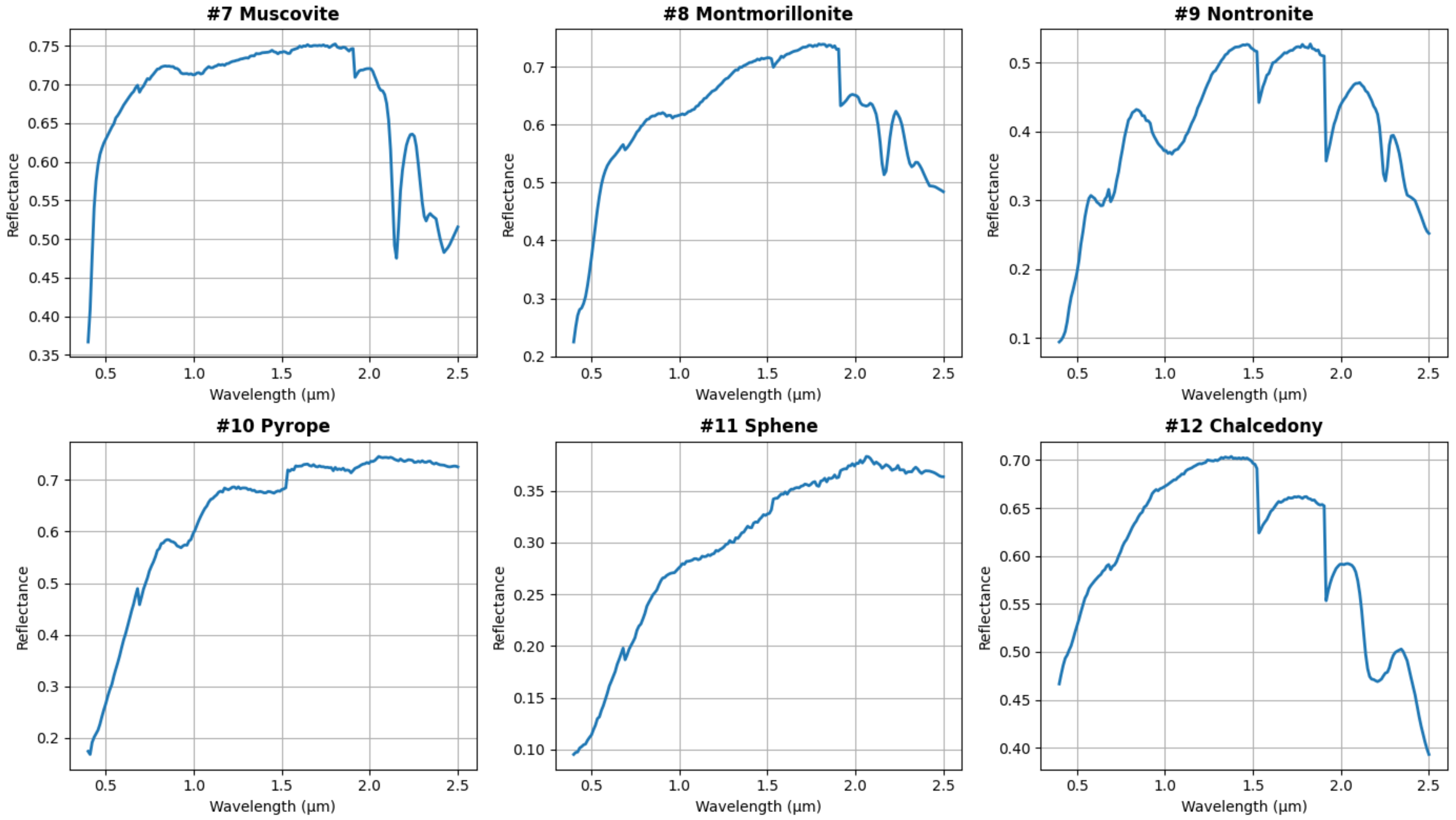
Real Dataset — AVIRIS Cuprite with FCLS-Derived Abundances [6]



Endmembers (12 minerals — USGS Spectral Library)



Endmembers (12 minerals — USGS Spectral Library)



FCLS Abundance Generation

Fully Constrained Least Squares (FCLS) decomposes each pixel into fractional mineral contributions:

Objective:

$$\min_{\mathbf{a}} \quad \| \mathbf{E}^T \mathbf{a} - \mathbf{p} \|^2$$

Subject to:

$$a_k \geq 0 \ \forall k \ \textit{(non-negativity)}$$

$$\sum_{k=1}^{12} a_k = 1 \ \textit{(sum-to-one)}$$

Where:

$\mathbf{p} \in \mathbb{R}^{188}$ = observed pixel spectrum

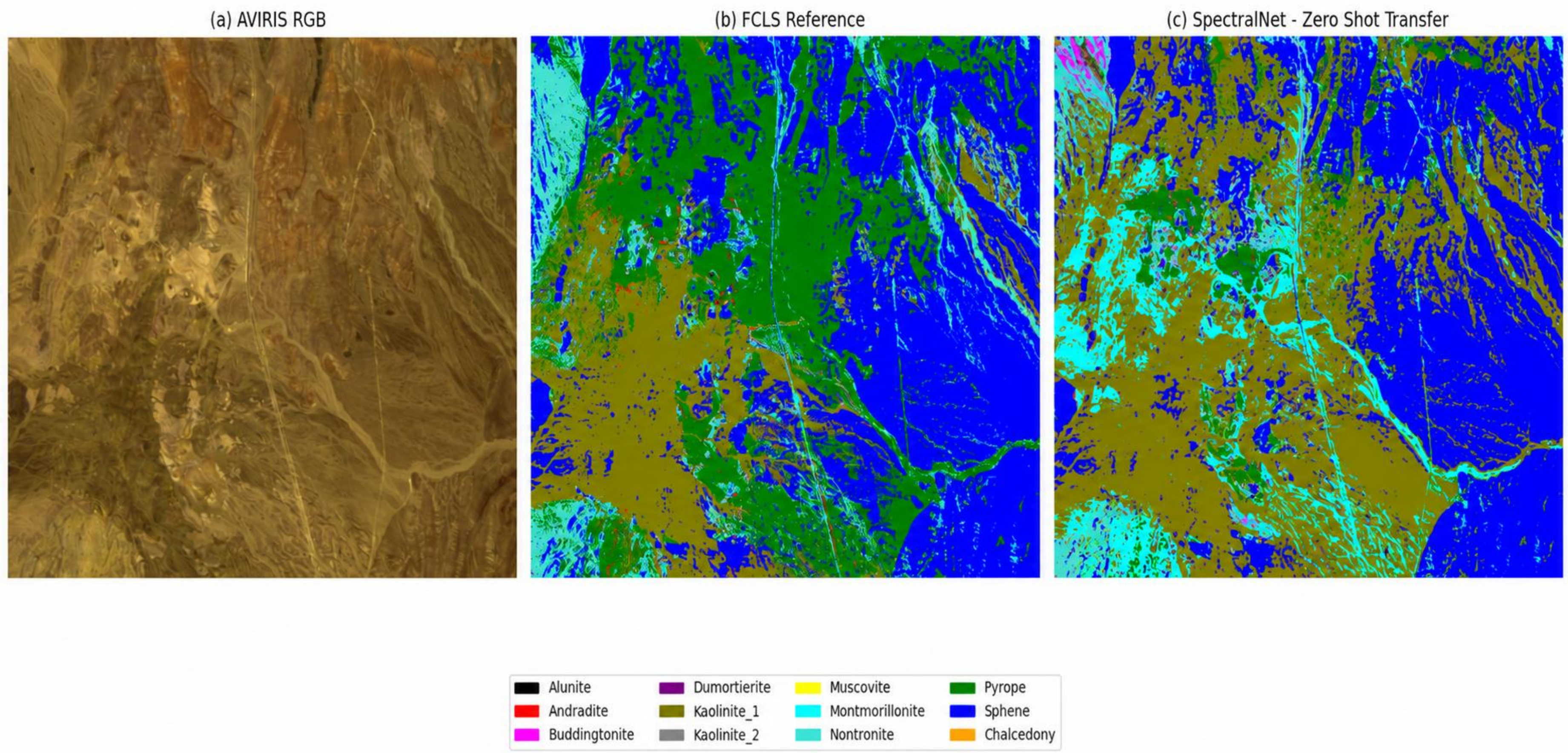
$\mathbf{E} \in \mathbb{R}^{12x188}$ = endmember matrix (12 minerals × 188 bands)

$\mathbf{a} \in \mathbb{R}^{12}$ = fractional abundance vector (output)

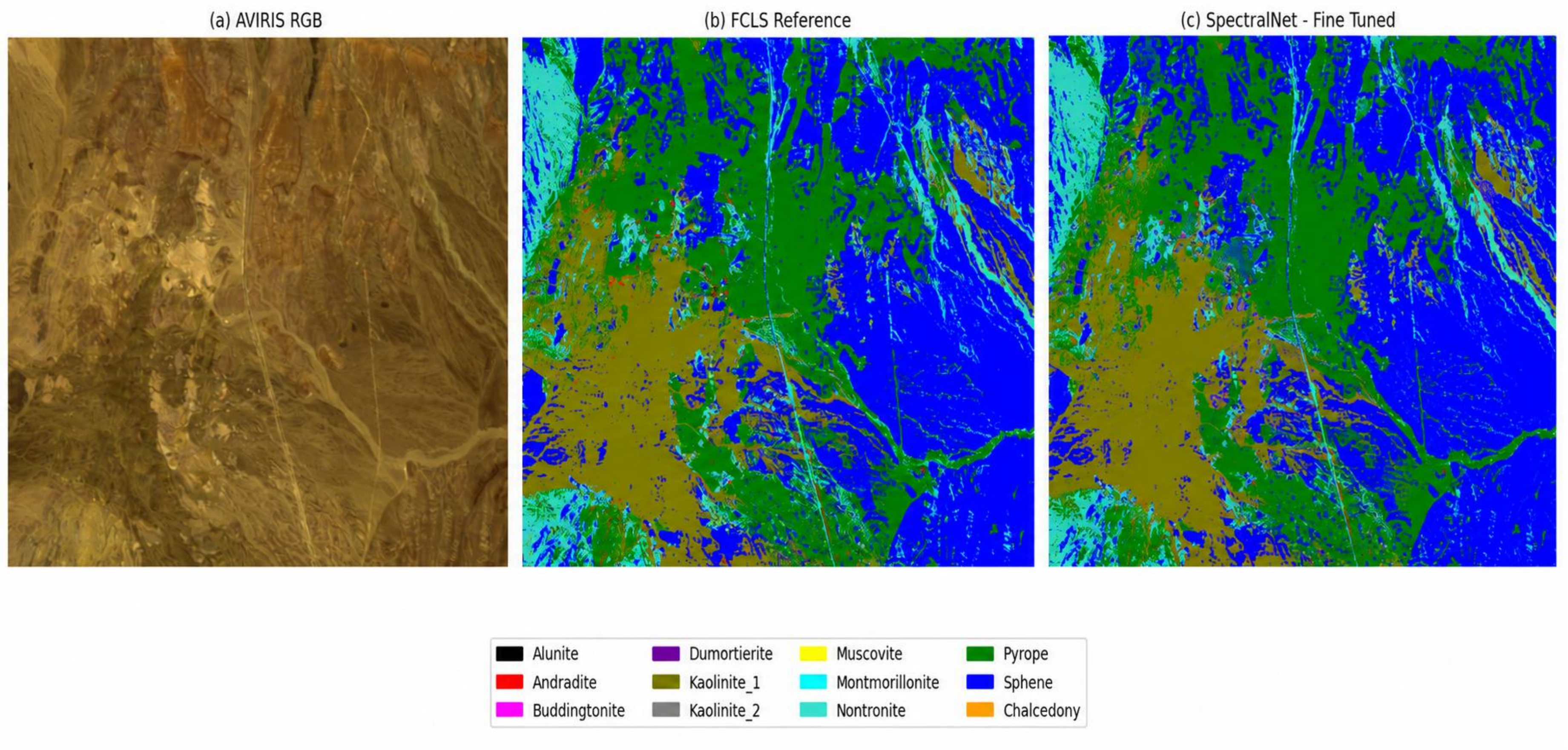
Results - Quantitative

Model	**Dataset**	**Protocol**	**NRMSE ↓**	**Overall Accuracy ↑**	**Kappa ↑**
DUENet (Pattathal et al. 2022)	Real AVIRIS Cuprite	Random Split	0.12	97.56%	0.95
SpectralNet	Synthetic Cuprite	Random Split	0.064	99.08%	0.9898
SpectralNet	Real AVIRIS Cuprite	Zero-shot transfer	0.163	54.02%	0.40
SpectralNet	Real AVIRIS Cuprite	Fine-tuned	0.1227	88.60%	0.8378

Results – Qualitative (Zero Shot Transfer)



Results – Qualitative (Fine Tuned Model)



Conclusion

- SpectralNet achieved high accuracy on synthetic Cuprite data and demonstrated effective transfer to real AVIRIS Cuprite imagery.

- DTW-based spectral similarity and Gaussian interpolation improved spectral feature learning and representation.

- Linear capsule projection reduced routing complexity while maintaining strong discriminative performance.

- The framework generated spatially consistent abundance and mineral classification maps with efficient convergence.

References

[1] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, *"Convolutional Autoencoder for Spectral–Spatial Hyperspectral Unmixing."*

[2] Y. Su, J. Li, A. Marinoni, A. Plaza, P. Gamba, and S. Chakravortty, *"DAEN: Deep Autoencoder Networks for Hyperspectral Unmixing,"* IEEE Transactions on Geoscience and Remote Sensing, vol. 57, no. 7, 2019.

[3] S. Shi, M. Zhao, L. Zhang, and J. Chen, *"Variational Autoencoders for Hyperspectral Unmixing with Endmember Variability,"* IEEE ICASSP, 2021.

[4]Pattathal V., Sahoo, Porwal, and Karnieli,
"Deep-learning-based Latent Space Encoding for Spectral Unmixing of Geological Materials."

[5] K. Mantripragada and F. Z. Qureshi, "Hyperspectral Pixel Unmixing with Latent Dirichlet Variational Autoencoder," IEEE Transactions on Geoscience and Remote Sensing, vol. 62, pp. 1–13, Jan. 2024, doi: 10.1109/TGRS.2024.3357589.

[6] Dataset Source:
[https://www.ehu.eus/ccwintco/uploads/7/7d/Cuprite_f970619t01p02_r02_sc03.a.rfl.mat]

Thank You

Gargee Pandey